



MOBILE APPLICATION FOR DIRECT MARKET ACCESS FOR FARMERS

A MINI PROJECT REPORT

Submitted by

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In partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND BUSINESS SYSTEMS

V.S.B ENGINEERING COLLEGE, KARUR.

(An Autonomous Institution)

ANNA UNIVERSITY: CHENNAI - 600025

JULY 2026

BONAFIDE CERTIFICATE

Certified that this project report "MOBILE APPLICATION FOR DIRECT MARKET ACCESS FOR FARMERS" is the bonafide work of "KAAVIYA A (922525244023)" who carried out the project work under my supervision.

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DECLARATION

I declare that the project report titled "MOBILE APPLICATION FOR DIRECT MARKET ACCESS FOR FARMERS" is the result of original work done by me and, to the best of my knowledge, similar work has not been submitted to "ANNA UNIVERSITY, CHENNAI" for the requirement of the Degree of Bachelor of Engineering. This project report is submitted in partial fulfilment of the requirement of the award of the Degree of Bachelor of Engineering.

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ACKNOWLEDGEMENT

First and foremost, I express my thanks to my parents for providing me with a very nice environment for doing this project. I wish to express my sincere thanks to our Founder and Chairman Shri. V.S.BALSAMY B.Sc., B.L., for his endeavor in educating us in this premier institution.

I wish to express my appreciation and gratefulness to our Principal, Dr.C.VENNILA M.E., Ph.D., and Vice Principal Dr.T.S.KIRUBASANKAR M.E., Ph.D for their encouragement and sincere guidance.

I am grateful to our Head of the Department Dr.V.JAMUNA M.E., Ph.D., and our project coordinator for their valuable support throughout the course of this work.

I express my sincere thanks to the supervisor of my project [Supervisor Name], M.E.,(Ph.D), Assistant Professor, Department of Computer Science and Business Systems, for his/her guidance throughout the course of my project.

My sincere thanks to all the teaching staff of V.S.B Engineering College and my friends for their help in the successful completion of this project work.

Finally, I bow before God, the Almighty who always had a better plan for me. I give my praise and glory to Almighty God for the successful completion of this project.

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Abstract

Indian agriculture continues to be dominated by long, multi-tiered supply chains in which farmers sell their produce to local intermediaries and commission agents rather than directly to buyers. This dependency on middlemen significantly reduces the price realised by farmers, delays payments, and leaves farmers with little visibility into fair, real-time market prices. The proposed Mobile Application for Direct Market Access, named KisanDirect, is a mobile-friendly web application developed using HTML5, CSS3, and Vanilla JavaScript that connects farmers directly with buyers, eliminating unnecessary intermediaries from the transaction chain. The application allows farmers to create verified profiles, list their crops with quantity and expected price, and track buyer orders through a simple dashboard. Buyers can browse live crop listings, view current market prices for various commodities, and place orders directly with farmers. The system incorporates Aadhaar-based farmer verification and UPI payment references to build trust and transparency between the two parties. A dedicated market price tracking module keeps farmers informed of prevailing rates so that they can negotiate fairly. This project demonstrates how a lightweight, accessible mobile web application can meaningfully improve price realisation, reduce transaction delays, and promote transparency in agricultural trade, particularly for small and marginal farmers with limited digital literacy.

Mapped Program Outcomes (POs): PO1 - Engineering Knowledge, PO2 - Problem Analysis, PO3 - Design/Development of Solutions, PO5 - Modern Tool Usage, PO7 - Environment and Sustainability, PO9 - Individual and Team Work, PO12 - Life-long Learning.

OBJECTIVES

The Mobile Application for Direct Market Access for Farmers has been designed with the following key objectives:

- To develop a mobile-friendly platform that connects farmers directly with buyers, eliminating intermediaries.
- To enable farmers to create verified profiles and list crops with quantity, quality, and expected price.
- To provide buyers with a simple interface to browse crop listings and place orders directly with farmers.
- To display real-time or regularly updated market prices for major agricultural commodities.
- To incorporate Aadhaar-based verification and UPI payment references for trust and transparency.
- To maintain an order management system that tracks the status of transactions from listing to sale.
- To improve price realisation for farmers and reduce dependency on commission agents and middlemen.

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LIST OF ABBREVIATIONS

S.NO	ABBREVIATION	EXPANSION
1	HTML	HyperText Markup Language
2	CSS	Cascading Style Sheets
3	JS	JavaScript
4	UI	User Interface
5	UX	User Experience
6	UPI	Unified Payments Interface
7	API	Application Programming Interface
8	DB	Database
9	KYC	Know Your Customer
10	PDF	Portable Document Format

CHAPTER 1

INTRODUCTION

1.1 Introduction

Agriculture remains the backbone of the Indian economy, yet a large share of the value generated in the farm-to-market chain is captured by intermediaries rather than the farmers who grow the produce. Farmers typically sell their harvest to local commission agents or middlemen at the nearest mandi, often at prices well below the prevailing market rate, due to limited access to buyers, lack of real-time price information, and dependency on the same set of local traders season after season.

The Mobile Application for Direct Market Access, KisanDirect, is a mobile-friendly web-based platform that allows farmers to list their crops directly for sale and connect with buyers without going through intermediaries. The application provides farmers with a simple interface to create verified profiles, list available produce with quantity and price, and track incoming orders. Buyers, in turn, can browse verified crop listings, view live market price references, and place orders directly with the farmer.

The main objective of this project is to develop a lightweight, accessible, and mobile-first application that improves farmers' price realisation, shortens the supply chain, and brings transparency to agricultural trade, while remaining simple enough for farmers with limited digital literacy to use effectively.

1.2 Existing Manual Practices and Their Limitations

Currently, most small and marginal farmers sell their produce through local mandis or commission agents. These intermediaries typically decide the price with limited transparency, deduct commissions before payment, and may delay payment by several days or weeks. Farmers rarely have access to real-time price data from other markets, which puts them at a significant disadvantage during price negotiation. Furthermore, the absence of any centralised, verified platform makes it difficult for buyers outside the farmer's immediate locality to discover and purchase produce directly.

1.3 Need for a Direct Market Access System

As mobile and internet penetration grows in rural India, there is a significant opportunity to reduce farmers' dependency on intermediaries by giving them direct, verified access to buyers. A digital direct market access system minimises the number of intermediaries in the supply chain, gives farmers visibility into real-time market prices, and speeds up payment cycles through digital references such as UPI. Such a system also benefits buyers by giving them access to fresher produce at more competitive prices, while supporting the broader goal of improving farmer incomes and reducing post-harvest losses caused by delayed sales.

1.4 Objectives and Scope

- To develop a mobile-friendly platform enabling farmers to list crops directly for sale.
- To allow buyers to browse verified listings and place orders directly with farmers.
- To provide a market price tracking module showing prevailing rates for major commodities.
- To incorporate Aadhaar-based farmer verification and UPI payment references for trust.
- To maintain an order management system tracking transactions from listing to sale.
- To improve price transparency and reduce farmers' dependency on middlemen.

The scope of this project is limited to a web-based mobile-friendly prototype covering crop listing, order management, market price display, and farmer/buyer profiles, with the design allowing future extension to a full backend, native mobile application, and payment gateway integration.

1.5 Applications

- Individual farmers seeking direct buyers for their harvest.
- Farmer Producer Organisations (FPOs) coordinating group sales.
- Wholesale buyers, retailers, and food processing companies sourcing produce directly.
- Government and NGO-led agricultural market linkage initiatives.
- Rural digital literacy and e-governance programs.

1.6 Advantages

- Improves price realisation for farmers by removing unnecessary intermediaries.
- Provides real-time visibility into market prices to support fair negotiation.
- Speeds up payment cycles through digital, UPI-based references.
- Builds trust between farmers and buyers through Aadhaar-based verification.

1.7 Software and Hardware Requirements

Hardware Requirements

- Processor: Intel Core i3 or above / any smartphone processor
- RAM: 4 GB or higher
- Storage: 250 GB Hard Disk/SSD (development machine)
- Display: 1366 x 768 resolution (development), any mobile screen (usage)
- Internet Connection (required for live data and order updates)

Software Requirements

- Operating System: Windows 10/11 / Android / iOS
- Programming Language: HTML5, CSS3, JavaScript (Vanilla JS)
- IDE: Visual Studio Code
- Browser: Google Chrome or Microsoft Edge (mobile and desktop)

1.8 Chapter Summary

This chapter introduced the motivation, problem statement, objectives, applications, advantages, and technical requirements of the Mobile Application for Direct Market Access for Farmers. The following chapters describe the related literature, the detailed system design and methodology, the implementation carried out, the results obtained through testing, and the conclusions drawn from this project along with its future scope.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Direct market access for farmers has been studied extensively in the context of agricultural supply chain reform, e-governance, and mobile commerce. Researchers and government bodies have proposed various approaches ranging from physical farmer markets to digital e-mandi platforms. This chapter reviews existing studies, government initiatives, and limitations of current agricultural market linkage systems.

2.2 Existing System

Traditionally, farmers sell produce through local mandis regulated under the Agricultural Produce Market Committee (APMC) system, where commission agents facilitate the sale in exchange for a fee. National-level electronic platforms such as the e-NAM (National Agriculture Market) portal have attempted to digitise mandi trading, but adoption among small and marginal farmers remains limited due to low digital literacy, dependency on local agents for logistics, and limited smartphone-friendly interfaces.

2.3 Literature Review

Study 1: E-NAM and Digital Mandi Platforms

Government-led studies on the e-NAM initiative found that while electronic trading portals improved price discovery at the mandi level, farmer-level adoption remained low because the platforms were designed primarily for traders and required registration through APMC yards rather than direct farmer-to-buyer interaction.

Study 2: Mobile-Based Agricultural Market Linkage Apps

Research into mobile applications for farmer-buyer linkage found that simple, lightweight interfaces with local language support significantly increased adoption among rural users compared to feature-heavy platforms designed for urban audiences.

Study 3: Digital Identity Verification in Rural Platforms

Studies on trust-building in peer-to-peer agricultural platforms recommended government ID-based verification, such as Aadhaar, to reduce fraud and build confidence between previously unacquainted farmers and buyers transacting directly for the first time.

Study 4: Digital Payments in Agricultural Transactions

Research into UPI adoption in rural India showed that digital payment references considerably reduced payment delays and disputes compared to cash-based settlement, which was previously the norm in mandi transactions.

Study 5: Market Price Transparency and Farmer Bargaining Power

Economic studies on price transparency found that farmers with access to real-time market price information negotiated significantly better prices with buyers and intermediaries, supporting the case for embedding a market price reference module directly within a farmer-facing application.

2.4 Comparative Analysis

Compared to traditional mandi-based selling and government e-mandi portals, the proposed KisanDirect application focuses specifically on direct farmer-to-buyer interaction with a lightweight, mobile-first interface, built-in market price references, and simple identity/payment verification, rather than replicating the trader-oriented workflow of existing e-mandi systems.

Feature	Traditional Mandi / e-NAM	Proposed System (KisanDirect)
Selling Route	Through commission agents	Direct farmer-to-buyer
Price Visibility	Limited, agent-controlled	Live market price reference module
Verification	Minimal / manual	Aadhaar-based farmer verification
Payment Speed	Often delayed, cash-based	UPI reference, faster settlement
Target User	Traders, large registered farmers	Small and marginal farmers

CHAPTER 3

SYSTEM DESIGN / METHODOLOGY

3.1 Introduction

System design defines how the Mobile Application for Direct Market Access for Farmers is organised and how each component interacts with the others. The proposed system allows farmers to register, list crops, and manage orders, while buyers browse listings, view market prices, and place orders directly.

3.2 System Architecture

The system consists of the following components:

- Farmer Profile and Verification Module
- Buyer Interface and Order Placement Module
- Crop Listing Module
- Market Price Tracking Module
- Order Management Module
- Shared Design System (UI components, styling)

When a farmer lists a crop, the listing is published to the buyer-facing catalogue. A buyer can then browse listings, check current market prices for reference, and place an order. The order management module updates both farmer and buyer dashboards with the current status of the transaction.

3.3 Technologies Used

- Frontend: HTML5, CSS3, Vanilla JavaScript
- Structure: Multi-file folder layout with a shared design system
- Verification Reference: Aadhaar number field with format validation
- Payment Reference: UPI ID field for transaction reference
- Browser Storage: Client-side storage for prototype data persistence

3.4 Module Description

Module 1: Farmer Module

- Create and manage a verified farmer profile.
- List crops with quantity, quality grade, and expected price.
- View and manage incoming buyer orders.

Module 2: Buyer Module

- Browse available crop listings by category and location.
- View current market price references for comparison.
- Place orders directly with farmers.

Module 3: Market Price Module

- Display prevailing market prices for major commodities.
- Help farmers set fair listing prices and negotiate confidently.

Module 4: Order Management Module

- Track order status from placement to completion.
- Maintain a history of past transactions for both farmers and buyers.

3.5 Data Storage Design

Field Name	Description
farmer_id	Unique identifier for each farmer
farmer_name	Name of the registered farmer
aadhaar_ref	Aadhaar-based verification reference
crop_name	Name of the listed crop
quantity	Available quantity of the crop
expected_price	Farmer's expected price per unit
buyer_id	Unique identifier for each buyer
order_status	Current status of the order
upi_ref	UPI payment reference for the transaction

3.6 Data Flow

The data flows through the following stages: Farmer Registration -> Crop Listing -> Buyer Browsing -> Order Placement -> Order Status Update -> Payment Reference -> Transaction Completion.

3.7 Non-Functional Requirements

- Performance: Crop listings and market prices should load within one to two seconds on a standard mobile connection.
- Reliability: The order status should accurately reflect the true state of each transaction.
- Usability: The interface should be simple enough for farmers with limited digital literacy.
- Scalability: The design should support a growing number of farmer and buyer profiles.
- Portability: The application should run on any modern mobile or desktop browser.

3.8 Planned Feature Enhancements

- Add a Node.js/MongoDB backend for persistent, multi-device data storage.

- Integrate Razorpay for real payment processing instead of UPI reference fields.
- Develop a React Native native mobile application for wider accessibility.
- Introduce AI-based crop price prediction to help farmers time their sales.
- Add regional language support for wider adoption among farmers.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Implementation Overview

The Mobile Application for Direct Market Access, KisanDirect, was implemented as a multi-file HTML5, CSS3, and Vanilla JavaScript prototype with a structured folder layout and a shared design system across all pages. The prototype covers farmer registration and verification, crop listing, buyer browsing, market price tracking, and order management.

4.2 Sample Output - Market Listings Dashboard

The buyer-facing dashboard displays a catalogue of active crop listings along with the farmer's location, listed quantity, expected price, and a reference to the current market price for the same commodity, allowing buyers to quickly judge whether a listing is fairly priced.

4.3 Feature-wise Evaluation

- Farmer registration and Aadhaar-based verification functioned correctly across test profiles.
- Crop listings were correctly published and made visible to buyers in real time within the prototype.
- The market price tracking module displayed reference prices alongside farmer listings as intended.
- Order placement and status updates correctly reflected each stage of the transaction.

4.4 Discussion

Testing of the prototype confirmed that a simple, mobile-first interface can effectively support direct farmer-to-buyer transactions without the complexity of a full trading platform. Market price references alongside listings helped simulate more informed price negotiation, and the Aadhaar/UPI reference fields demonstrated how identity and payment trust signals could be embedded directly into the listing and order flow.

4.5 Advantages Achieved

- Removes dependency on intermediaries by connecting farmers directly with buyers.
- Improves price transparency through an embedded market price reference module.
- Builds trust between parties through Aadhaar-based verification and UPI references.
- Provides a simple, mobile-friendly interface suitable for low digital literacy users.

4.6 Comparison with Traditional Mandi Selling

Aspect	Traditional Mandi Selling	Proposed System (KisanDirect)
Number of Intermediaries	One or more commission agents	None - direct farmer-to-buyer
Price Transparency	Low, agent-dependent	Market price reference shown to both parties
Payment Speed	Often delayed	UPI reference speeds up settlement
Farmer Effort	Physical travel to mandi	Listing from mobile device

4.7 Limitations of the Current System

- The current prototype uses client-side storage and does not yet have a persistent backend database.
- Payment is handled via a UPI reference field rather than an actual integrated payment gateway.
- Market price data is manually maintained rather than fetched from a live external market price API.
- The system does not yet include logistics or delivery coordination between farmer and buyer.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

Reducing farmers' dependency on intermediaries and improving price transparency are pressing needs in Indian agriculture as digital access continues to expand in rural areas. The Mobile Application for Direct Market Access, KisanDirect, was developed as a lightweight, mobile-friendly platform that connects farmers directly with buyers, incorporating crop listings, order management, market price tracking, and Aadhaar/UPI-based trust signals.

The application successfully demonstrates how a simple HTML5, CSS3, and Vanilla JavaScript prototype can model a meaningful improvement to the agricultural supply chain, reducing farmers' dependency on middlemen and giving them greater visibility into fair market prices. This project shows that even a lightweight, accessible web application can meaningfully improve price realisation and transaction transparency for small and marginal farmers.

5.2 Future Enhancements

- Integrate a Node.js/MongoDB backend for persistent, multi-device data storage.
- Integrate Razorpay or a similar gateway for real, in-app payment processing.
- Develop a React Native native mobile application for wider rural accessibility.
- Introduce AI-based crop price prediction to help farmers choose the best time to sell.
- Add regional language support and voice-based navigation for low-literacy users.
- Integrate logistics and delivery coordination between farmers and buyers.

5.3 Overall Outcome

The Mobile Application for Direct Market Access for Farmers successfully achieves the project objectives by providing a functional, mobile-first prototype that connects farmers directly with buyers. The application is easy to use, lightweight, and suitable for both educational demonstration and as a foundation for a full-scale agricultural market linkage platform.

APPENDIX A: GLOSSARY OF TERMS

Term	Meaning
Direct Market Access	Selling produce directly to buyers without intermediaries.
Aadhaar Verification	Identity verification using India's national ID system.
UPI	A real-time digital payment system widely used in India.
Crop Listing	A farmer's published entry describing a crop available for sale.
Market Price Reference	The prevailing rate for a commodity shown to inform pricing decisions.
Order Management	The process of tracking a transaction from placement to completion.

APPENDIX B: SAMPLE NOTIFICATION MESSAGE

When a buyer places an order, the farmer receives a notification similar to the following, giving clear context and next steps:

"New Order Received: A buyer has placed an order for 50 kg of Tomatoes at your listed price of Rs.20/kg. Please confirm the order to proceed with the sale."

This concise message format was designed to convey the necessary information without overwhelming farmers with limited digital literacy, consistent with the notification design principles discussed in the literature survey.

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